

Testing Guide — AI Virtual Building Tour

Step-by-step tests for the local MVP. Run each case, compare to the expected result, and mark Pass / Fail. Cases you don't run should be marked **Not Executed** — not assumed Pass.

A · Setup & test data

App URL	<code>http://localhost:8000</code> (start with <code>./run.sh</code>)
Login	<code>demo@vbt.local</code> / <code>demo1234</code> (role: <code>company_admin</code> , email pre-verified)
Sample plan	Upload <code>virtual-building-tour/sample-floorplan.png</code> (a 2-bed flat drawing made for testing)
API docs	<code>http://localhost:8000/docs</code> — try every endpoint interactively

Note on AI detection: OpenCV is optional. If it's **not** installed, AI analysis returns a built-in template layout (still lets you test the full flow). To test **real** detection on the sample image, run `pip install opencv-python-headless numpy` in the venv and restart.

B · Happy path (main flow)

ID	STEP	EXPECTED RESULT	P / F
T1	Open app → log in with demo account	Dashboard loads; header shows "Demo Designer · company_admin"	
T2	Create project: name, type = Apartment, location	Opens project wizard; status pill = Draft	
T3	Building details: floors=1, height=3 → Save building	Toast "Building saved"; values persist on reload	
T4	Click + Add floor	"Floors: L0" appears under the form	
T5	Upload → choose sample-floorplan.png → Upload	Toast "Uploaded"; filename shown; status → Files Uploaded	
T6	AI analysis → Run AI analysis	Toast "Detected N walls, M rooms"; 2D preview canvas + warnings appear	
T7	2D Verify: check scale = 0.02 → Approve 2D plan	Toast "Plan approved"; status → Approved for 3D	
T8	3D · Tour · Video → Generate 3D building	"Model ready" link to a .glb ; status → 3D Ready	
T9	Click Build tour , then Open 3D viewer	New tab shows 3D building; you can orbit/zoom; scene list on left	
T10	In viewer → ▶ Play walkthrough	Camera flies room-to-room through the scenes	
T11	Back on project → ratio 16:9 → Render video	"completed" + a download-MP4 link (needs FFmpeg); status → Completed	

C · Negative & edge cases

ID	STEP	EXPECTED RESULT	P/F
T12	Log in with a wrong password	Red error "Invalid email or password" (generic, no detail leak)	
T13	Upload a .txt or .docx file	Rejected: "Unsupported format" (only jpg/png/pdf allowed)	
T14	Try Generate 3D before approving a plan	Error "Approve a 2D plan first"	
T15	Try Run AI analysis before uploading a file	Error "Upload a floor plan first"	
T16	Register a NEW account, then log in and try analysis	Blocked "Verify your email..." until verified (local mode auto-verifies on register)	
T17	Reload the browser mid-wizard	Still logged in; project data persists (autosave)	
T18	Open viewer with a bad token: /viewer.html?token=xyz	"Could not load tour" message, no crash	

D · 3D viewer checks (BRD 16 interactive tour)

ID	CHECK	EXPECTED	P/F
T19	Building geometry	White walls, floor slab, brown door / glass window markers; empty rooms (no furniture)	
T20	Controls	Left-drag orbit · scroll zoom · right-drag pan · Reset view re-centres	
T21	Click a scene in the left list	Camera flies smoothly to that room	
T22	Multi-floor: set floors=2 in building, regenerate	Two stacked floors visible in the model	

E · API / admin testing

ID	STEP	EXPECTED	P/F
T23	Open /docs , run POST /api/auth/login	Returns an access_token ; use "Authorize" to call others	
T24	GET /api/admin/overview	Project counts by status, job counts, users	
T25	GET /api/admin/audit	Audit trail: login, create_project, approve_plan, etc.	
T26	GET /api/notifications	"review required", "3D model ready", "render completed"	

Automated test: a full pipeline test already passes 21/21 (register → upload → analyse → approve → 3D → tour → render). The cases above are the manual UI equivalents — use them to confirm the browser experience.

Mark each case Pass / Fail / Not Executed. File any Fail with the case ID, steps, and what you saw vs. expected. Disclaimer (BRD): outputs are visual representations only — not certified documents.